

Top separation sequences for LDPE/EVOH/PET

Dynamic-programming candidates at 100 C, ranked by minimum step selectivity

■ Good

■ Marginal

1

LDPE -> EVOH -> PET

min selectivity **11.9%**

Step	Target	Solvent	Sel.
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1	LDPE	p-Xylene	11.9%
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2	EVOH	DMSO	18.4%
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S1 opt: 89.6% @150 C | S2 opt: 44.9% @160 C

Good selectivity window

2

EVOH -> LDPE -> PET

min selectivity **11.9%**

Step	Target	Solvent	Sel.
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1	EVOH	DMSO	18.4%
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2	LDPE	p-Xylene	11.9%
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S1 opt: 37.6% @145 C | S2 opt: 89.6% @150 C

Good selectivity window

3

EVOH -> PET -> LDPE

min selectivity **5.0%**

Step	Target	Solvent	Sel.
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1	EVOH	DMSO	18.4%
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2	PET	DMF	5.0%
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S1 opt: 37.6% @145 C

Marginal selectivity window

Solvent labels: p-Xylene = 1,4-dimethylbenzene; DMSO = dimethylsulfoxide; DMF = dimethylformamide. Sel. is step selectivity at 100 C.